

TOPICS

1. ADVERSE DRUG REACTION
2. PHYTOTHERAPY
3. DRUG DEVELOPMENT

ANTENEH KASSAHUN(M.D, MSC)

ADVERSE DRUG REACTIONS (ADR)

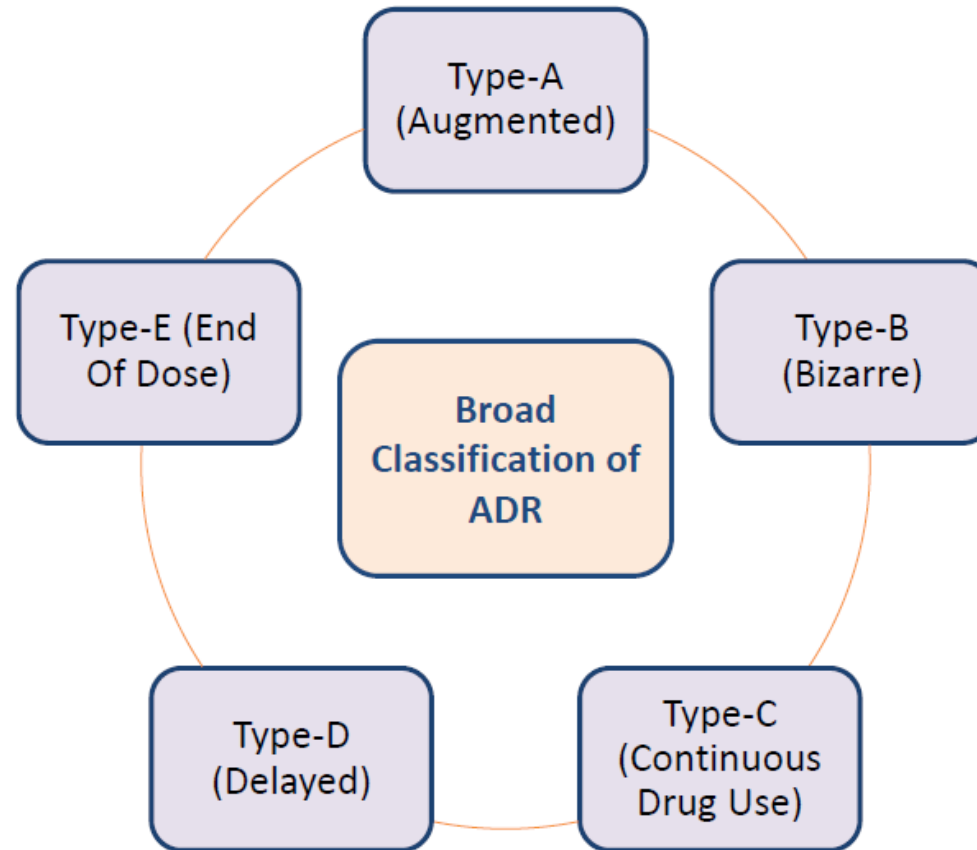
❖ A noxious and unintended response to a medicine used in humans for prophylaxis, diagnosis, or therapy



Factors in ADR

- Age
- Sex
- Genetics
- Comorbidities
- Polypharmacy
- Dose
- Duration

CLASSIFICATION OF ADR



- Adverse drug reactions are claimed to be the fourth leading cause of death in USA.
- Some adverse reactions, such as side effect, overdose may occur in anyone.
- Adverse reactions occurring only in susceptible patients include idiosyncrasy (frequently genetic in origin), and allergy (usually immunologically mediated due to genetic predisposition)

TYPE-A(AUGMENTED)

Commonest

Dose dependent, severity increases with dose

Predictable by pharmacological mechanism

Preventable in most part by slow introduction of low dosages

i) **Side effects**-unwanted effect

E.g.- dry mouth by TCA

- diarrhea with antibiotics

- respiratory dep with opioids

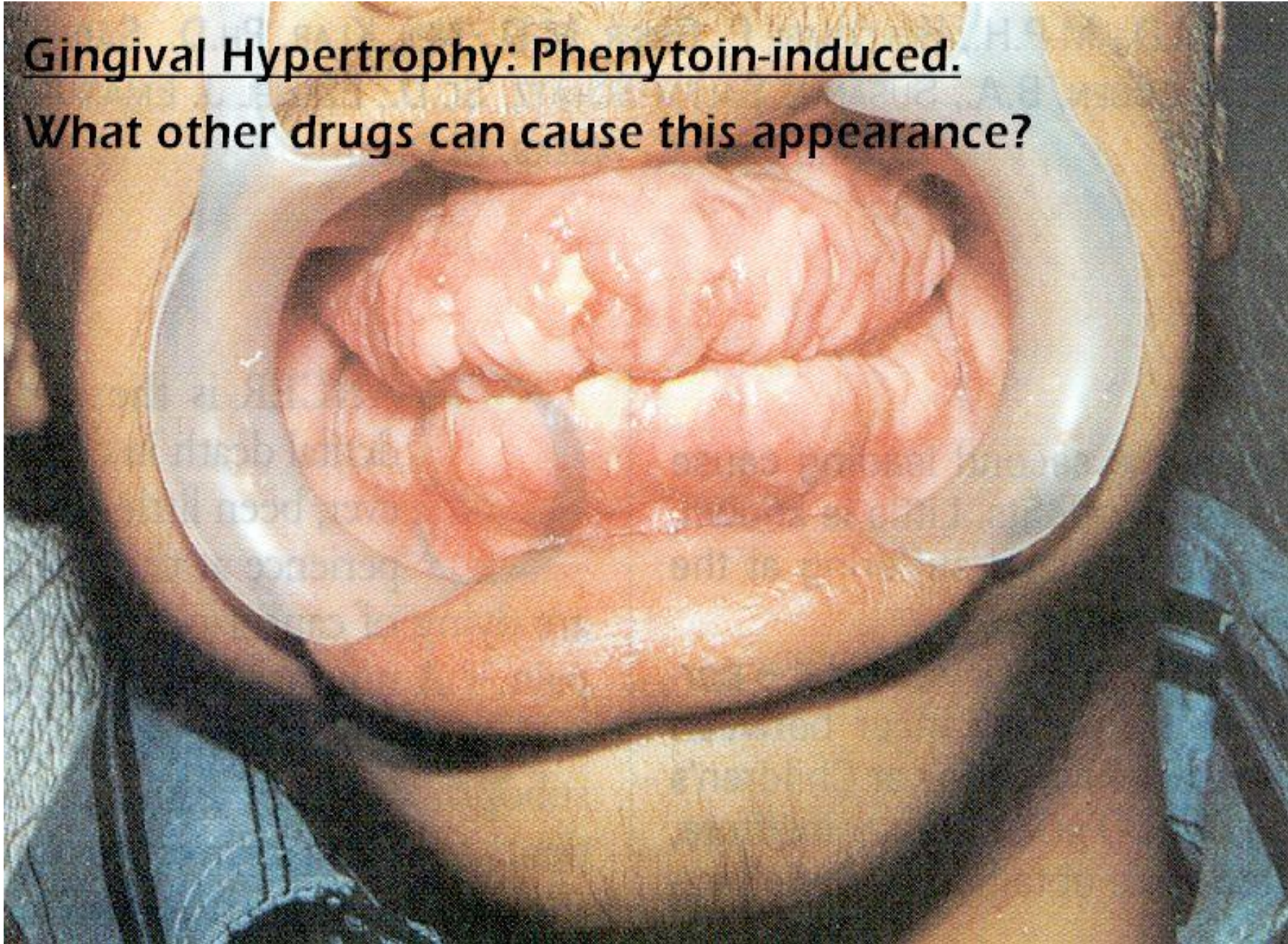
- hypoglycemia by insulin

•ii) **Toxicity/overdose**- excessive pharmacological effect

- E.g – bleeding with Heparin, hepatic injury with acetaminophen

Gingival Hypertrophy: Phenytoin-induced.

What other drugs can cause this appearance?



TYPE B(BIZARRE)



- Rare
 - Unpredictable and unrelated to the dose
 - Unrelated from the drug's known mechanism of action
- i) **Idiosyncratic Rxn**- abnormal rxn to drug which is genetically determined, Mechanism are unknown, serious, can be fatal
- E.g.- hemolytic anemia by primaquine in G6PD def
- peripheral neuropathy by isoniazid
- ii) **Hypersensitivity Rxn**-
- **E.g** anaphylaxis with penicillin

Hypersensitivity reaction/allergy/ :

- ❖ ***Type I(IgE mediated):*** antigen + IgE on mast cell.. immediate degranulation & release of histamine and tryptase
 - E.g- Anaphylaxis, asthma, hay fever or urticaria
- ❖ ***Type II:*** antigen + IgM/IgG
 - E.g –penicillin induced hemolytic anemia
 - -Transfusion reactions
- ❖ ***Type III (immune complex mediated reaction):*** mediated mainly by IgG
 - E.g-glomerulonephritis, SLE
- ❖ ***Type IV (cell mediated reaction) :*** delayed hypersensitivity reaction- CD8+ cytotoxic T cells kill targeted cells.
 - E.g- contact dermatitis(poison ivy), SJS
 - PPD for TB infection

Steven's Johnson syndrome(SJS)



TYPE-C (CONTINUOUS OR CHRONIC DRUG USE)

Dose related

Uncommon

Result of continuous drug use(cumulative dose)

May be irreversible

E.g. - Osteoporosis with corticosteroids

-Nephrotoxicity with NSAID

- Retinopathy with Ethambutol

TYPE-D (DELAYED)

Delayed occurrence of ADRs after cessation of treatment

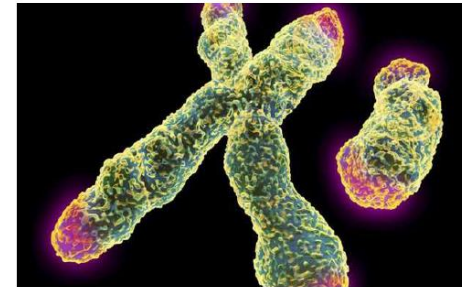
Time related

Uncommon

i) **Carcinogenesis**- chr abnormality...Ca

- hormonal Rx, anticancer drugs, RT

ii) **Teratogenesis**- fetal abnormality if taken in 1st TMP(e.g thalidomide)



Teratogenicity

Table I-2-2. FDA Classification of Drugs and Pregnancy

Category	Risk	
	Animals	Humans
A	–	–
B	+/–	–/o
C	+/o	o
D	+	+
X	+	+

- A: very safe
- B: safe
- C: may be safe
- D & X: teratogenic

– = Studies have proven absense of teratogenicity;
o = no studies available;
+ = studies have proven teratogenicity

TYPE-E (END OF THERAPY)

- Withdrawal reactions soon after cessation of the drug

E.g.-

- adrenal crisis on cessation of corticosteroids
- Rebound insomnia, seizures on benzodiazepines withdrawal

Rx- reintroduce the drug and withdraw slowly





TYPE F (failed therapy)

- Underdose
- Drug interaction
- Counterfeit

MANAGEMENT OF ADRS

1. If the reaction is mild

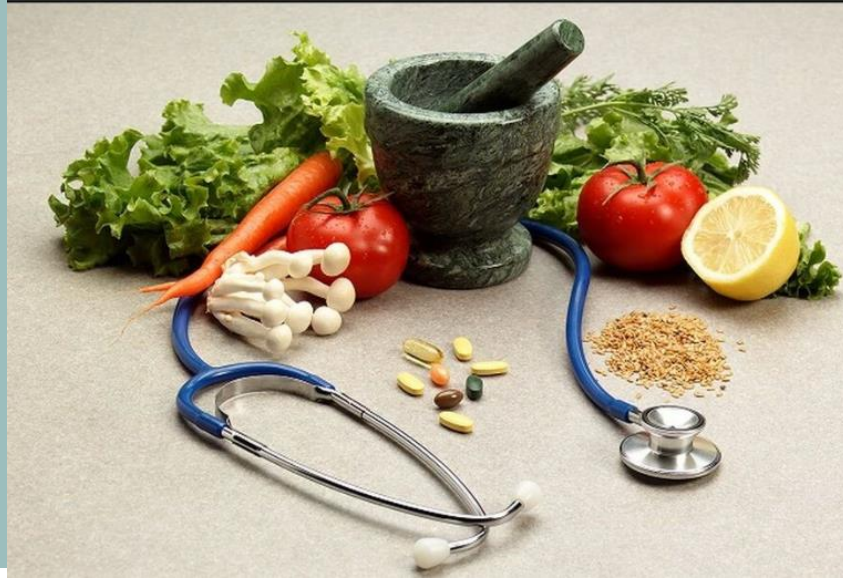
- Continue treatment if necessary
- Stop unnecessary drugs
- Consider dose reduction
- Symptomatic treatment if warranted.

2. If the reaction is serious

- Withdraw suspected drugs
- Urgent treatment



PHYTOTHERAPY



❖ The production of medicines and the pharmacologic treatment of diseases began with the use of herbs.

❖ Traditional medicine

❖ Conventional/modern medicine



- ❖ Phytotherapy is a field of medicine that uses plants either to treat disease or as health-promoting agents.
- ❖ It is often referred to as herbalism in Western medicine.
- ❖ Traditional use of phytotherapies generally uses original composition and integrity of the source plant for medicinal purposes.
- ❖ The more traditional use of phytotherapy often includes the whole part of the plant, such as a chamomile herb infusion (tea), whereas Western herbal medicine more commonly uses single herb part and standardized to a component of the extract.
- ❖ Several medical traditions use plant-based therapies, including anthroposophic medicine, naturopathic medicine, traditional Chinese medicine (TCM), Ayurvedic medicine.

- ❖ Pharmaceutical medications derived from plants are typically single compounds isolated through the industrial separation and extraction of components identified as having therapeutic properties

COMMON HERBAL MEDICINES

- ❖ Ginseng, garlic, and echinacea are among the 10 most common natural products used among adults in western countries.

Garlic(*Allium sativum*)

- ❖ is associated in popular culture with its use as a treatment of the common cold and, although a trial does exist that suggests its efficacy in prevention of common cold occurrences, more evidence is necessary to confirm this.
- ❖ Some studies suggest it lowers blood pressure, possess antibiotic and antiviral properties and may be useful in the treatment of cancer, but more studies are needed.

Ginseng (*Panax ginseng*)

- ❖ has been shown to have a positive effect on cognition, memory, quality of life, and behavior, though further studies are needed for confirmation.
- ❖ Its positive effects on well-being, appetite, and sleep have made it a promising target for addressing quality of life, especially in patients with cancer, multiple sclerosis, and other chronic illnesses.
- ❖ Ginseng seems to exert anti-inflammatory effects through its chemical ginsenosides.



PLATE 14:
Ginseng
(*Panax ginseng*)



PLATE 12:
Ginger
(*Zingiber officinale*)



Garlic

Echinacea

- ❖ Echinacea is a plant primarily used to treat upper respiratory infections.
- ❖ A recent Cochrane review has shown that, in general, echinacea has some preventive effects in terms of the common cold.
- ❖ Proposed mechanism of echinacea is immunomodulatory, including the enhancement of macrophage function.

Turmeric (*curcumin longa*)

- ❖ Component- curcumin
- ❖ Ant-inflammatory and anti-oxidant effect
- ❖ Might have use for - joint pain
 - Ulcerative colitis



PLATE 29:
Turmeric



PLATE 9:
Echinacea
(*Echinacea purpurea*)

St. John's wort(*Hypericum perforatum*)

- ❖ Can grow to 60 cm, yellow flowers
- ❖ Grows wild everywhere
- ❖ One of the most popular and well-known phytomedicines
- ❖ Components- hypericin, pseudo-hypericin
- ❖ Proposed Mechanism- inhibit reuptake of serotonin,
Dopamine and NE
- ❖ In some places used as anti-depressant,
- ❖ Has many drug-drug interactions



PLATE 26:
St John's wort
(*Hypericum perforatum*)

Traditional medicine

- TM has been employed since times immemorial
- Till the 19th century almost all medical practice were traditional
- TM is affordable and accessible
- Drawback-Several active ingredient and often unknown

Conventional(Western) medicine

- are pure chemicals
- prescribed by doctors
- expensive

TRADITIONAL MEDICINE AND DRUG DISCOVERY

- 80% of people in developing countries utilize plants to meet their primary health care needs
- Ethiopia: The national 1982-83 rural health survey revealed that more than half of all health seekers used traditional healers or self-treatment.
- China and India:
 - Integrated traditional medicine
 - Formal training system

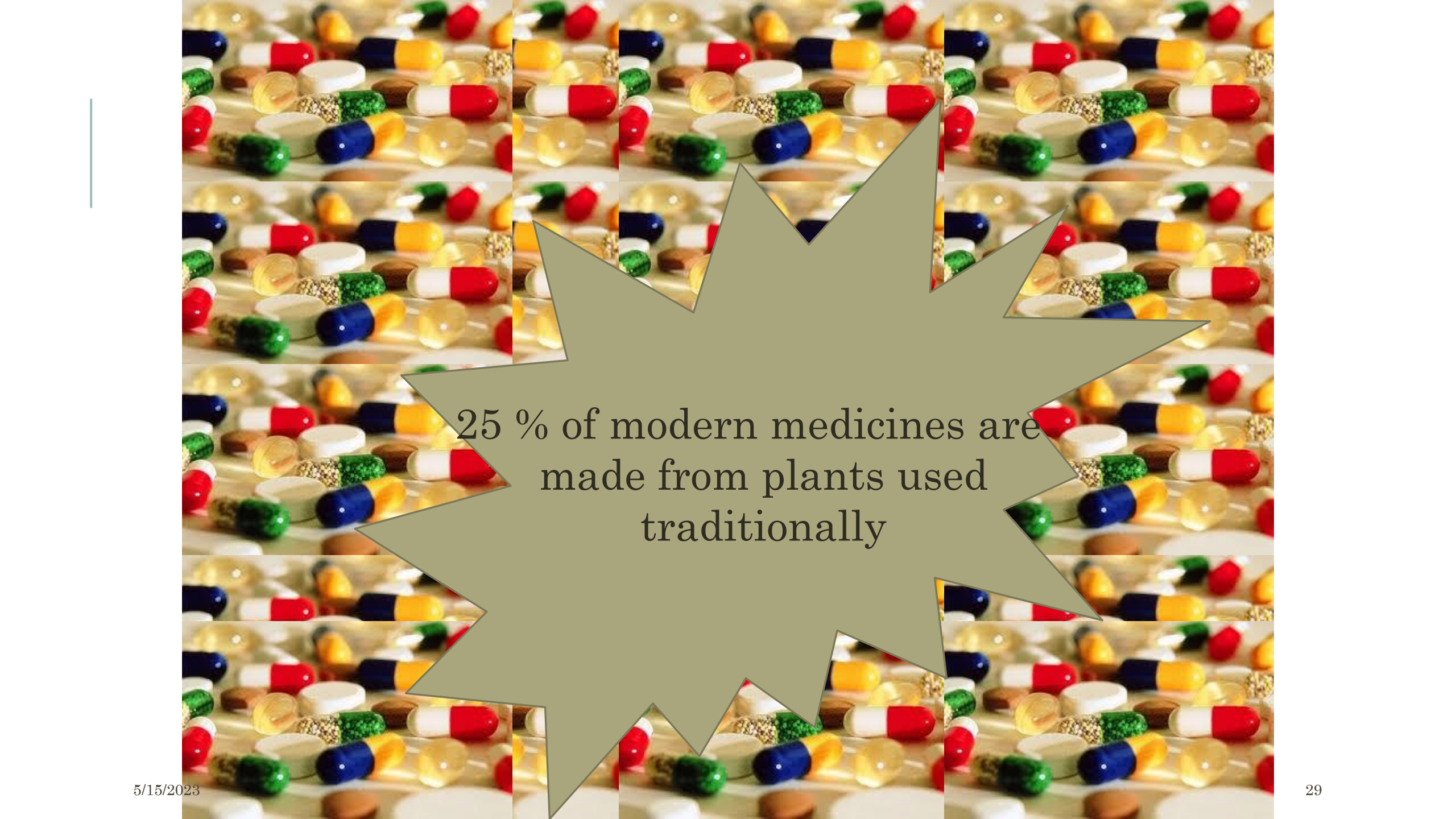
- ❖ Plant extracts are now rarely used by physicians although herbal remedies are still popular with the public (1 in 5 adults in the USA report using a herbal product and annual cost of herbs in USA is 9 bil \$ in 2019)
- ❖ Even if the health professional doesn't accept the validity of these older medical system, it is necessary to know about their basic principles

Examples of plant constituents that are isolated for medicinal use. Naturally, these constituents do not occur alone in plants but as fractions accompanied by related chemical compounds. The isolated substances, which generally have strong, immediate actions, are not considered phytomedicines in the strict sense.

These are called plant-derived drugs.

Constituent/Drug	Plant	Action
Atropine	Belladonna leaves.& roots	Parasympatholytic
Caffeine	Coffee shrub	Analeptic, CNS stimulant
Cocaine	Coca leaves	Local anesthetic
Colchicine	Autumn crocus bulb	Gout remedy, antiinflammatory
Digoxin	Digitalis/Foxglove leaves	Cardiac remedy (positive inotropic)
Emetine	Ipecac root	Emetic
Ephedrine	Ephedra herb	Antihypotensive
Kawain	Kava root/.rhizome	Anxiolytic
Morphine	Opium poppy exudate	Analgesic
Physostigmine	Calabar bean	Cholinesterase inhibitor
Pilocarpine	Jaborandi leaves	Glaucoma remedy
Quinidine	Cinchona bark	Antiarrhythmic
Quinine	Cinchona bark	Antimalarial
Reserpine	Rauwolfia/Indian snakeroot	Antihypertensive
Salicin	Willow bark	Anti-inflammatory
Scopolamine	Datura spp./Jimson weed	Antispasmodic
Taxol®	Pacific yew bark	Cytostatic
Theophylline	Tea leaves	Bronchodilator

A significant portion of all currently used medications are derived, either directly or indirectly, from active principles that have been isolated from plants. Some well-known examples are listed in the this Table



25 % of modern medicines are
made from plants used
traditionally

Why is the use of herbal products increased?

- Increased use of dietary and herbal weight loss supplements for obesity
- For disease prevention
- The increasing cost of conventional medicines
- Traditional belief that herbs are safer and more effective than prescription drugs

PLANT-DERIVED DRUGS

Analgesics:

- Aspirin: *Salix* species/Europe
- Morphine, Codeine: *Papaver somniferum*/opium poppy/ (latin Americas)

Cardiotonic:

- Digitalis: *Digitalis purpurea*/UK-Europe

Malaria:

- Quinine: *Cinchona* spp./Amazon
- Artemisinin: *Artemisia annua*/China

Antihypertensive:

- Reserpine: *Rauwolfia serpentina*/India

Muscle relaxant:

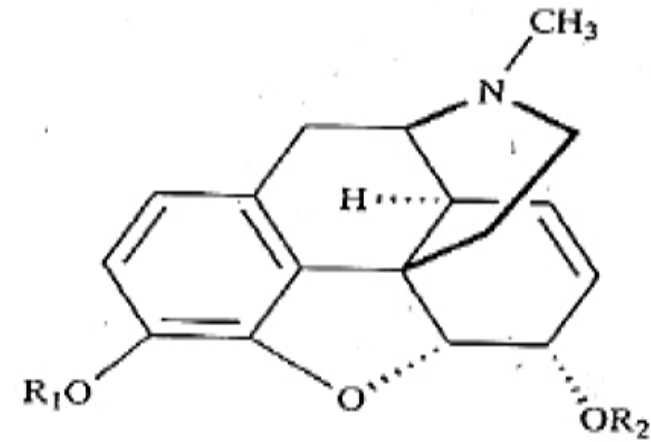
- Tubocurarine: *Chondrodendron* spp./Amazon



Fleurs et capsules du pavot
(*Papaver somniferum*,
Papaveraceae). Le latex blanc,
c'est l'opium.



Hostettmann, K. *Tout Savoir sur le Pouvoir des Plantes*. 1997:132



morphine $R_1 = R_2 = H$

codeine $R_1 = CH_3$, $R_2 = H$

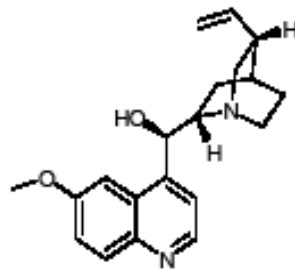
heroin $R_1 = R_2 = COCH_3$

Papaver somniferum

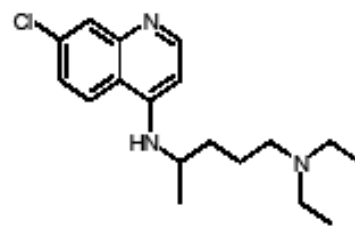
ANALGESICS



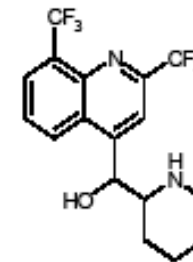
ANTIMALARIAL DRUGS



Quinine



Chloroquin



Mefloquine

Quinine from *Cinchona* spp. used in the Amazon region for centuries for treatment of fevers.

PLANT-DERIVED ANTICANCER DRUGS IN CLINICAL USE

- **Vinblastine/Vincristine** :- *Catharanthus roseus*/Jamaica, Philippines (originally from Madagascar)
- **Paclitaxel/Docetaxel**:- *Taxus spp*/NW US, Europe
- **Topotecan/Irinotecan**: *Camptotheca acuminata*/China
- **Etoposide**:- *Podophyllum spp*/ Eastern US, Himalayas

IDENTIFICATION OF POTENTIAL MEDICINAL PLANT

- **Ethnobotany/** Ethnopharmacological study
- **Medico-religious manuscripts** and traditional pharmacopoeia
- **Taxonomist/** pharmaceutical botanist
- **Traditional healer**

Sources of knowledge for traditional healers (N=44), Addis Ababa and Butajira, Central, Ethiopia

Sources	Number	Percent
Family	30	68.2
Religious Institutions	5	11.4
Preceding sickness & corresponding use	3	6.8
Apprenticeship	3	6.8
Gift of god	2	4.5
Family & Preceding sickness	1	2.3

(Gedif and Hahn, 2002)

Number of plant ingredients in traditional prescriptions (N=277), Addis Ababa and Butajira, Central Ethiopia

Composition of prescriptions	Number of prescriptions (and number of species involved)	Percent
Single species	247 (247)	89.2
double species	20 (40)	7.2
Triple species	7 (21)	2.5
Quadruple species	3 (12)	1.1
Total	277 (320)	100.0

(Gedif and Hahn, 2002)

DRUG DISCOVERY AND DEVELOPMENT

- All drugs are toxic in human, what matters is.....
 - the dose.
- Sources of drugs
 - Plants
 - Animals
 - Minerals
 - Microorganisms
 - Synthetic
 - Genetic engineering

Drug discovery approaches:-

- Rational drug design: based on an understanding of drug receptor structure, and drug structure
- Chemical modification of a known molecule
- Screening for biologic activity of natural products and synthetic library
- Combinations of known drugs to obtain additive or synergistic effects or a repositioning of a known drug for a new therapeutic use.
- Serendipity

- **Traditional methods** of drug discovery rely on trial-and-error testing of chemical substances
- **Modern drug discovery** involves the identification of screening hits, medicinal chemistry and optimization of those hits to increase the
 - affinity,
 - selectivity (to reduce the potential side effects),
 - efficacy/potency

DRUG DISCOVERY/INVENTION AND DEVELOPMENT

- ❖ Drug development is the process of bringing a new pharmaceutical drug to the market once a lead compound has been identified through the process of drug discovery.
- ❖ Drug discovery is a lengthy, expensive, difficult, and inefficient process with low rate of new therapeutic discovery.
- ❖ Drug discovery and development comprises
 1. Basic research and lead discovery (discovery or synthesis of a potential new drug compound)
 2. Preclinical development/study
 3. Clinical development/study

BASIC RESEARCH AND LEAD DISCOVERY

Identifying a Drug Target

- Drug Target = specific macromolecule, or biological system, which the drug will interact with
- Sometimes this can happen through incidental observation...



Screening chemical library or natural sources



Hits (5k to 10k)



Leads (<20).....



preclinical study



Clinical trial



New Drug (1)

Drug discovery

Drug Development

Why do we test hits and leads?

- Hits and leads -have promising activity against a particular biological target that is important in disease.
- However, little is known about the safety, toxicity, pharmacokinetics and PD of these chemicals in humans.
- It is the function of drug developer to assess all of these parameters prior to human clinical trials.
- Thousands of compounds may be synthesized and hundreds of thousands tested from libraries of compounds for each successful new drug lead, which then frequently needs to be further optimized for reasons of potency, selectivity, drug metabolism, and dosing convenience before each drug reaches the market.

BIOASSAY

In vitro



In vivo



FINDING THE LEAD



Screening Natural Products

- Plants, microbes, the marine world, and animals
- Assay
- Structure elucidation

FINDING THE LEAD (CONT.)



Screening synthetic banks

- Pharmaceutical companies have prepared thousands of compounds
- These are stored (in the freezer), cataloged and screened on new targets

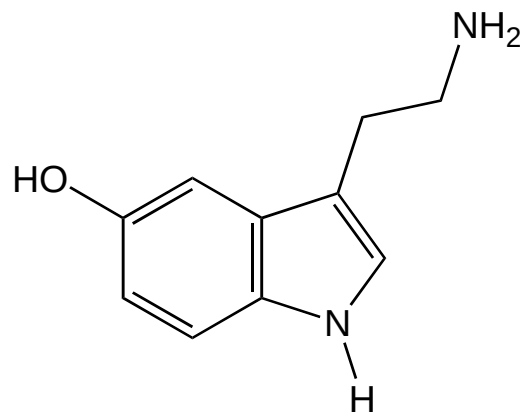
FINDING THE LEAD (CONT.)



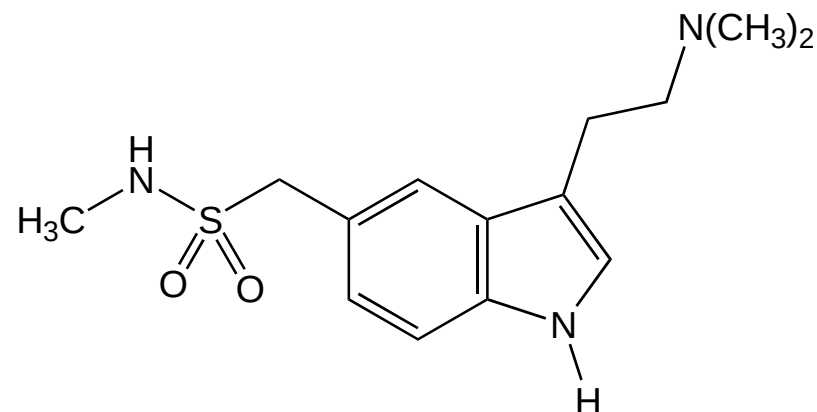
Using Someone Else's Lead(chem optimization)

- Design structure which is similar to existing lead, but different enough to avoid patent restrictions.
- Sometimes this can lead to dramatic improvements in biological activity and pharmacokinetic profile.

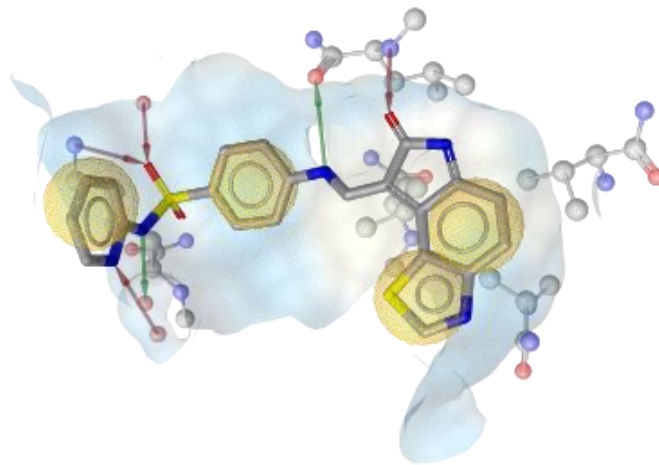
Use structural similarity to a natural ligand



5-Hydroxytryptamine (5-HT)
Serotonin (a natural neurotransmitter
synthesized in certain neurons in the CNS)



Sumatriptan (Imitrex)
Used to treat migraine headaches
known to be a 5-HT₁ agonist



Computer-Assisted Drug Design

- If one knows the precise molecular structure of the target (enzyme or receptor), then one can use a computer to design a perfectly-fitting ligand.

Serendipity: a chance occurrence

Example:

- **Penicillin** discovery (in 1928 by Fleming)
- Development of **Viagra** (1998 by Pfizer company)



Lead molecule is identified then what?

- After bioassay studies and optimization, it will be studied in vivo.
- For PK/PD study
- Toxicity

PRECLINICAL STUDY

- The goals of preclinical toxicity studies include
 - To identify potential human toxicities
- Toxicity testing is time-consuming and expensive. May need 2 to 6 years
- Large numbers of animals may be needed to obtain valid pre Clinical data
- Acute, subacute & chronic toxicity testing in animals, including carcinogenicity studies will be conducted
- Therapeutic index and toxicity data from animals to humans will be extrapolated
- But most drug candidates fail in this phase and don't reach to the market

PRECLINICAL STUDY (Cont'd)

- Pharmacodynamic, pharmacokinetic and toxicity will be studied
- Each animal is slightly different, with different metabolic systems, etc.
- Thus a drug may be toxic to one species and not to another



5/15/2020



CLINICAL TRIALS

- After animal study IND will be applied to FDA, if approved CT starts
- Safety and efficacy of a compound is studied in human
- Less than one third of the drugs tested in clinical trials reach the marketplace
- It often requires 4–6 years of clinical testing to accumulate and analyze all required data.

CLINICAL TRIALS

Phase I:

- Drug is tested on healthy volunteers to determine *toxicity* relative to dose and to screen for unexpected side effects
- Designed to test tolerable dose and prevent severe toxicity



- In (20-100) healthy volunteers

CLINICAL TRIALS

Phase II:

- Tested on 100-500 patients,
- To determine drug's efficacy
- Randomized and controlled
- has highest rate of drug failures(75%)



CLINICAL TRIALS

Phase III:

- Drug is tested on much larger group of patients and compared with existing treatments and/or with a placebo (??)
- Involves randomized controlled trial (in thousands of pts)
- To further establish and confirm safety and efficacy)
- Very expensive phase
- After completion, marketing approval requires submission of a New Drug Application (NDA) to FDA
- Full report of all preclinical and clinical data to FDA
 - approval or rejection



CLINICAL TRIALS

Phase IV:

- Drug is placed on the *market* and patients are monitored for side effects
- Report about safety & efficacy
- Renewal or cancellation of the product depends on comment of review

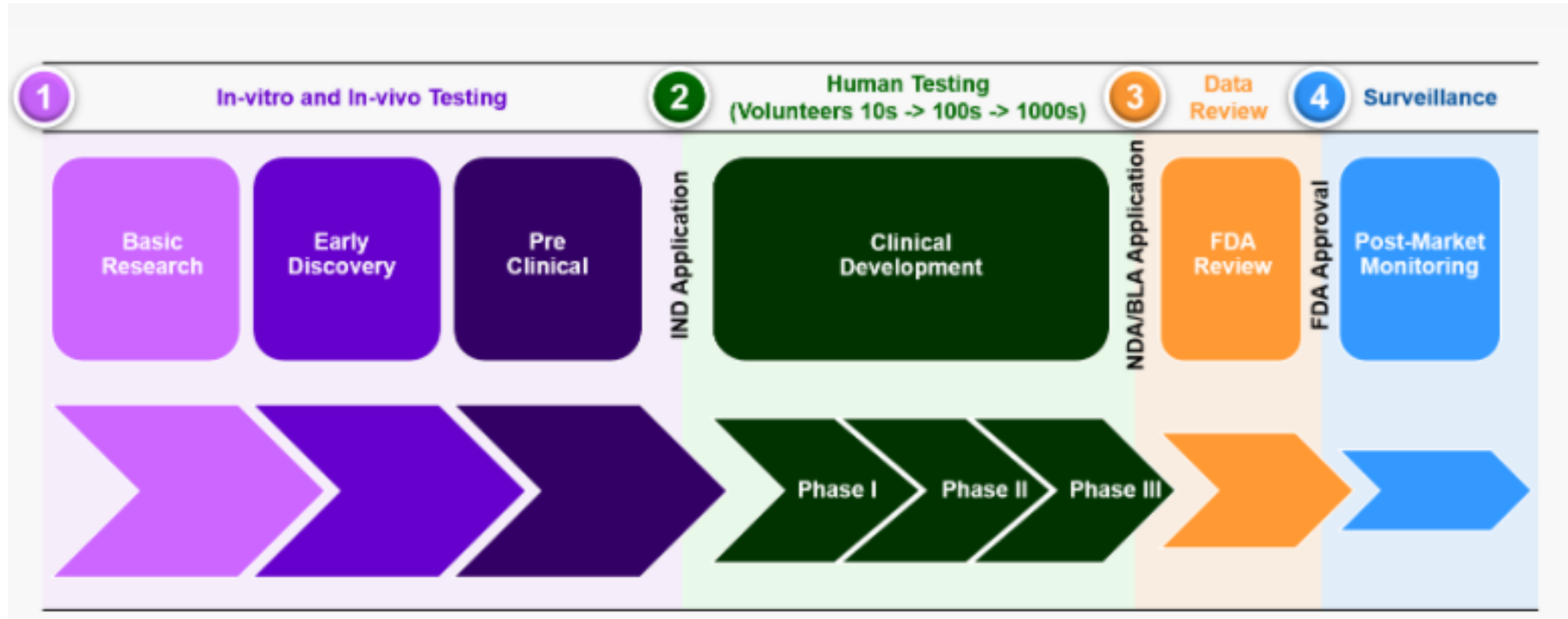


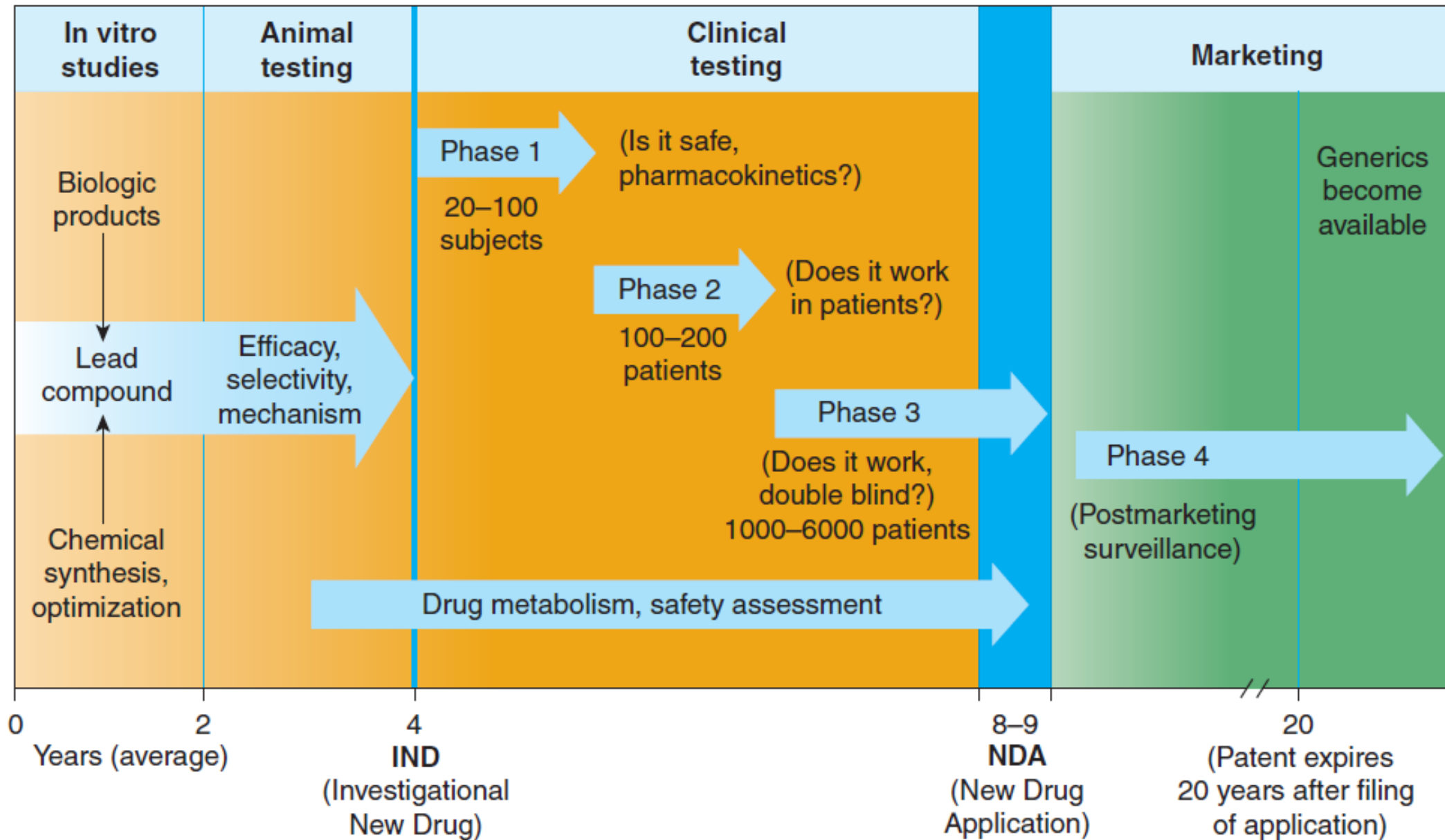
Birth defects caused by use of thalidomide



10,000 cases

SUMMARY: DRUG DISCOVERY & DEVELOPMENT





Cost of developing a drug?

- An average of 1.6 billion dollars

The time to discover and develop the drug to the market?

- 10-15 years

Generic drugs

- lifetime of a patent is 20 years then after 20 yrs, any company may produce the drug, and can market the drug as a generic product.

